

LEVEL 2: JAVA

1. Write a program to create a basic calculator that can perform addition, subtraction, multiplication, and division. The program should ask for two numbers (floating point) and perform all the operations

Hint =>

a.

Create a variable number1 and number 2 and take user inputs.

b.

Perform Arithmetic Operations of addition, subtraction, multiplication and division and assign the result to a variable and finally print the result

I/P => number1, number2

O/P => The addition, subtraction, multiplication and division value of 2 numbers ___ and ___ is ___, ___, ___, and ___

```
import java.util.Scanner;

public class Calculator {
    public static void main(String[] args){
        Scanner myObj = new Scanner(System.in);
        System.out.println("Enter number1");
        float number1 = myObj.nextFloat();
        System.out.println("Enter number2(!=0)");
        float number2 = myObj.nextFloat();
        //addition of 2 numbers
        float sum = number1 + number2;
        //subtraction of 2 numbers
        float difference = number2 - number1;
        //multiplication of 2 numbers
        float product = number1 * number2;
        // division of 2 numbers
        float div = number1 / number2; // considered number2 !=0

        System.out.println("The addition, subtraction, multiplication and
division value of 2 numbers "+number1+
        " and "+number2+" is "+sum+", "+difference+", "+product+" and "+ div);
        myObj.close();
    }
}
```

2.

Write a program that takes the base and height to find area of a triangle in square inches and square centimeters

Hint => Area of a Triangle is $\frac{1}{2} * \text{base} * \text{height}$

I/P => base, height

O/P => Your Height in cm is ____ while in feet is ____ and inches is ____

```
import java.util.Scanner;

public class AreaOfTriangle {
    public static void main(String[] args){
        Scanner myObj = new Scanner(System.in);

        System.out.println("Base of Triangle");
        float base = myObj.nextFloat();

        System.out.println("Height of Triangle");
        float height = myObj.nextFloat();

        float area = (float) (0.5 * base * height);
        System.out.println("Area = "+area);
        myObj.close();
    }
}
```

3.

Write a program to find the side of the square whose parameter you read from user

Hint => Perimeter of Square is 4 times side

I/P => perimeter

O/P => The length of the side is ____ whose perimeter is ____

```
import java.util.Scanner;

public class SquarePerimeter {
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        System.out.println("Enter Side of Square");
        double side = myObj.nextDouble();
        double perimeter = 4 * side;
        System.out.println("The Perimeter of Square with side: "+side+" is "+perimeter);
        myObj.close();
    }
}
```

4.

Write a program to find the distance in yards and miles for the distance provided by user in feet

Hint => 1 mile = 1760 yards and 1 yard is 3 feet

I/P => distanceInFeet

O/P => Your Height in cm is ____ while in feet is ____ and inches is ____

```
import java.util.Scanner;

public class feetToYards {
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        System.out.println("Enter parameters in feets");
        double feet = myObj.nextDouble();
        //1 mile = 1760 yards and 1 yard is 3 feet
        double miles = (feet - feet%(1760*3))/(1760*3);
        double yards= (feet%1760*3)/3;
        System.out.println("Your Height in feet is "+feet+" while in miles is "+miles+" and yards is "+yards);
        myObj.close();
    }
}
```

5.

Write a program to input the unit price of an item and the quantity to be bought. Then, calculate the total price.

Hint => NA

I/P => unitPrice, quantity

O/P => The total purchase price is INR ____ if the quantity ____ and unit price is INR ____

```
// Source code is decompiled from a .class file using FernFlower decompiler.
import java.util.Scanner;

public class purchasePrice {
    public purchasePrice() {
    }

    public static void main(String[] var0) {
        Scanner var1 = new Scanner(System.in);
        System.out.println("Enter unit Price:");
        double var2 = var1.nextDouble();
        System.out.println("Enter Quantity:");
        int var4 = var1.nextInt();
        double var5 = var2 * (double)var4;
    }
}
```

```

        System.out.println("The total purchase price is INR" + var5 + "if the
quantity " + var4 + "and unit price is INR" + var2);
    }
}

```

6.

Write a program to take 2 numbers and print their quotient and reminder

Hint => Use division operator (/) for quotient and moduli operator (%) for reminder

I/P => number1, number2

O/P => The Quotient is ____ and Reminder is ____ of two number ____ and ____

```

// Source code is decompiled from a .class file using FernFlower decompiler.
import java.util.Scanner;

public class purchasePrice {
    public purchasePrice() {
    }

    public static void main(String[] var0) {
        Scanner var1 = new Scanner(System.in);
        System.out.println("Enter unit Price:");
        double var2 = var1.nextDouble();
        System.out.println("Enter Quantity:");
        int var4 = var1.nextInt();
        double var5 = var2 * (double)var4;
        System.out.println("The total purchase price is INR" + var5 + "if the
quantity " + var4 + "and unit price is INR" + var2);
    }
}

```

7.

Write an **IntOperation** program by taking a, b, and c as input values and print the following integer operations a + b * c, a * b + c, c + a / b, and a % b + c. Please also understand the precedence of the operators.

Hint =>

- | | |
|----|--|
| a. | Create variables a, b, c of int data type. |
| b. | Take user input for a, b, and c. |
| c. | Compute 3 integer operations and assign the result to a variable |
| d. | Finally, print the result and try to understand operator precedence. |

I/P => fee, discountPrecent

O/P => The results of Int Operations are —, —, and —

```
// Source code is decompiled from a .class file using FernFlower decompiler.
import java.util.Scanner;

public class purchasePrice {
    public purchasePrice() {
    }

    public static void main(String[] var0) {
        Scanner var1 = new Scanner(System.in);
        System.out.println("Enter unit Price:");
        double var2 = var1.nextDouble();
        System.out.println("Enter Quantity:");
        int var4 = var1.nextInt();
        double var5 = var2 * (double)var4;
        System.out.println("The total purchase price is INR" + var5 + "if the
quantity " + var4 + "and unit price is INR" + var2);
    }
}
```

8.

Similarly, write the **DoubleOpt** program by taking double values and doing the same operations.

```
import java.util.Scanner;

public class doubleOperation {
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        System.out.println("Enter numbers below:");
        double a = myObj.nextDouble();
        double b = myObj.nextDouble();
        double c = myObj.nextDouble();
        double oper1 = a + b * c;
        double oper2 = a * b + c;
        double oper3 = c + a / b;
        double oper4 = a % b + c;
        System.out.println("The results of Int Operations are "+oper1+",
"+oper2+", "+oper3+" and "+oper4);
        myObj.close();
    }
}
```